

WIRELESS-POWERED LED

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Wireless power is transferred from transmitter to receiver coil using inductive coupling principle. For wireless power transfer to work efficiently, frequency of the transmitter and the

receiver must be tuned to the same frequency. LC tank of transmitter side produces an oscillating magnetic field of the tuned frequency. LC tank of receiver side is same as the frequency of transmitter side, thus

achieving high efficiency.

This wireless-powered LED can be used in mobile phone battery chargers, electric tooth brushes and low-powered loads such as LEDs, SMPs, rotors and so on. The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the transmitter unit is shown in Fig. 2. It is built around 5V voltage regulator 7805 (IC1), NE555 timer (IC2) and a few other components.

Positive voltage regulator 7805's output provides regulated 5V for NE555 and BC547. NE555 is configured as an astable multivibrator for pulse generation. The capacitor charges through resistors R1 and R2.

Charging time $t_1 = 0.693(R1 + R2) \times C3$

Discharging time $t_2 = 0.693(R2) \times C3$

Frequency of oscillation $F = 1.44 / ((R1 + 2R2) \times C3) = 18.94\text{KHz}$

C4 is the bypass capacitor at control voltage pin 5 of NE555. T1 is the driver for power MOSFET IRF540 (T2). T2 drives LC tank circuit resonating at a frequency of $F = 1 / (2\pi\sqrt{LC})$

where, $L = L1 = 180\mu\text{H}$ and $C = C5 + C6 = 0.1\mu\text{F} + 0.1\mu\text{F} = 0.2\mu\text{F}$

Therefore tuned frequency F is around 26KHz. It produces an oscillating magnetic field around the coil at a frequency of 26KHz linked with LC tank circuit of the receiver.

Circuit diagram of the receiver unit is shown in Fig. 3. It is built around buck/boost regulator IC MC34063 and a few other components.

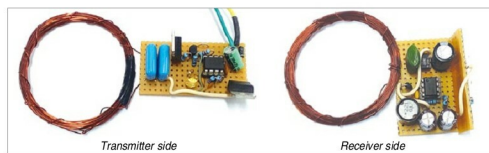


Fig. 1: Author's prototype

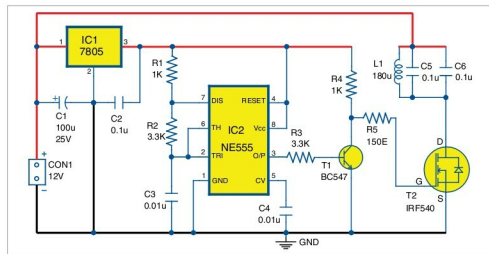


Fig. 2: Circuit diagram of the transmitter unit

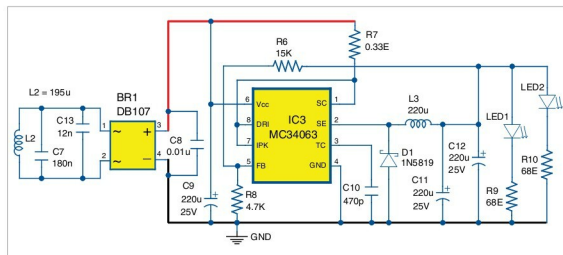


Fig. 3: Circuit diagram of the receiver unit